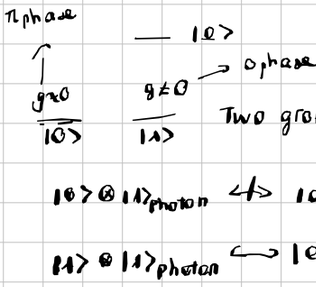
Duan Kimble Interaction

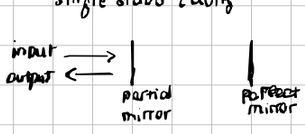
Duan Kimble interaction allows single photon to acquire a phase shift depending on quantum state of an atom inside a cavity



$|0\rangle \otimes |1\rangle_{photon} \leftrightarrow |0\rangle \otimes |0_{photon}\rangle$

$|1\rangle \otimes |1\rangle_{photon} \hookrightarrow |e\rangle \otimes |0\rangle_{photon}$

Reflected field consists of: incoming field that is to be reflected at mirror
field enters cavity and then leak



empty cavity

$H = \hbar\omega_c a^\dagger a$

incoming field injects a_{in} with rate $\sqrt{\kappa}$

$\dot{a} = (i\Delta_c - \frac{\kappa}{2})a + \sqrt{\kappa}\, a_{in}$ $\quad \Delta_c = \omega - \omega_c$

when state is steady

$\dot{a} = 0 \quad a = \frac{\sqrt{\kappa}}{\kappa/2 - i\Delta_c} a_{in} \quad a_{out} = a_{in} - \sqrt{\kappa}\, a = \left(1 - \frac{\kappa}{\kappa/2 - i\Delta_c}\right) a_{in}$

$r_0 = \frac{a_{out}}{a_{in}} \Rightarrow r_0(\omega) = 1 - \frac{\kappa}{\kappa/2 - i\Delta_c}$

on resonance $r_0 = -1$ or $e^{i\pi} \Rightarrow \phi = \pi$

when atom is coupled

$$r_1 = 1 - \frac{\kappa}{\kappa/2 - i\Delta_c + \frac{g^2}{\gamma/2 - i\Delta_a}}$$

For resonance and large cooperativity $\frac{g^2}{\gamma/2} \gg \kappa/2$

$r_1 \approx 1$

$= \frac{C-1}{C+1} \quad C = \frac{4g^2}{\kappa\gamma}$

So the idea is when it is not coupled, we have $r_0 \approx -1$ and when coupled $r_1 \approx 1$
atom in $|0\rangle$

That is when atom in $|0\rangle$, the cavity is single sided and acts as mirror creating a π phase shift by the atom
When coupled, atom and photo form two hybridized states

$$|\pm\rangle = \frac{|e0\rangle \pm |1,1\rangle}{\sqrt{2}} \qquad \omega_0 \pm g$$

the incoming is really not on the new resonances any more. $a \approx 0$

$a_{out} \approx a_{in}$

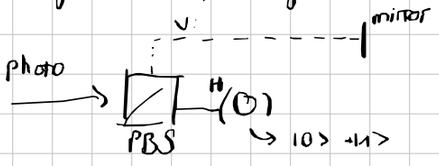
Consider atom is in state

$$|\psi_a\rangle = \alpha|0\rangle + \beta|1\rangle$$

Photon in

$$|\psi_p\rangle = c_H|H\rangle + c_V|V\rangle$$

H goes thr atom-cavity reflection, V goes thr reference path w/ no interaction



Photon and atom each have two states → 4 basis

Atom, photon				
$\|0, H\rangle$	-1	$\|0, V\rangle$	1
$\|1, H\rangle$	1	$\|1, V\rangle$	1

$\{|0,H\rangle, |0,V\rangle, |1,H\rangle, |1,V\rangle\}$

$U_{CR} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ controlled gate phase

$|\psi_{in}\rangle = \frac{1}{2}(|0,H\rangle + |0,V\rangle + |1,H\rangle + |1,V\rangle) \Rightarrow |\psi_{out}\rangle = \frac{1}{2}(-|0,H\rangle + |0,V\rangle + |1,H\rangle + |1,V\rangle)$

By defining

$$|-\rangle_P = \frac{|V\rangle - |H\rangle}{\sqrt{2}} \Rightarrow |\psi_{out}\rangle = \frac{1}{\sqrt{2}}(|0\rangle|-\rangle_P + |1\rangle|+\rangle_P)$$

ZALM

Imagine two ground stations, A and B separated by a large distance. We want to entangle their quantum memories without physically moving the memories.

Charlie: satellite quantum transmitter

& SPDC (χ^2) + Bell state analyzer gerate heralded entangled photons

$$|\psi^+\rangle_{AB} = \frac{|H_A V_B\rangle + |V_A H_B\rangle}{\sqrt{2}}$$

photon A → Alice

Mode convertor

↓

Cavity + spin A

↓

photon measurement

photon B → Bob

Mode coverter

↓

Cavity + spin B

↓

Photon measurement

Entangled Remote Quantum Memories

$$|\Phi^+\rangle_{A_S B_S} = \frac{|\downarrow_A \downarrow_B\rangle + |\uparrow_A \uparrow_B\rangle}{\sqrt{2}}$$

A and B each have an atom inside a cavity and they want to entangle the two atoms

Locally entangle each with a photon $|\psi\rangle_{Aa} = \frac{|0_A H_a\rangle + |1_A V_a\rangle}{\sqrt{2}}$ $|\psi\rangle_{Bb} = \frac{|0_B H_b\rangle + |1_B H_b\rangle}{}$

Charlie BSM performs a joint measurement on a and b $\Rightarrow |\psi\rangle_{ab} = \frac{|H_a V_b\sqrt{2} + |V_a H_b\rangle}{\sqrt{2}}$

Entanglement swap $\Rightarrow |\psi\rangle_{AB} = \frac{|0_A 1_B\rangle + |1_A 0_B\rangle}{\sqrt{2}}$. BSM relies on two photon interference at a BS

Hong Ou Mandel

Consider conventional route

A ⟶ (42, $\sqrt{\eta}$) ⟵ charlie ⟵ B

η

Transmission probability

$\eta(L) = e^{-\alpha L}$

$\eta_{half} = e^{-\alpha L/2} = (e^{-\alpha L})^{1/2}$

$= \sqrt{\eta}$

With transmission efficiency of η, for two photons to survive

$$P_{success} \propto \sqrt{\eta}\sqrt{\eta} = \eta$$

A and B are entangled and the send to charlie

Midpoint charlie creates photons and transfer to A and B. They then transfer that to their spin (Quantum)

A and B can retain information in memories while waiting the remote results This means that many trials can be used until success while the other succeeded photon gets stored

$$\bar{T}_{midpoint} \propto \sqrt{\eta}$$ rather than η

How coming photons entangle w/ V

Quantum interaction

$$r(\omega) = 1 - \frac{\kappa_{wg}(i\Delta_a + \gamma/2)}{(i\Delta_c + \kappa/2)(i\Delta_a + \gamma/2) + g^2}$$ ← wave guide coupling

When resonant, lossless and perfectly overcoupled

$r_\downarrow \approx 1$ coupled atom

$r_\uparrow \approx -1$ uncoupled

A real photon is a temporal wavepacket w/ a range of frequencies $|1_f\rangle = \int d\omega\, f(\omega)\, a^\dagger_{in}(\omega)|0\rangle$

$|\downarrow\rangle|1_g\rangle \rightarrow |\downarrow\rangle \int d\omega\, r_\downarrow(\omega) f(\omega)\, a^\dagger_{out}(\omega)|0\rangle + |\text{loss / env}\rangle$